0625 | PAPER 4

TOPIC 02
Thermal Physics

47 topical answers • 2021–2025
Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme. For drawings and graph lines, compare the stated method with the original question figure.

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Core formulas

- thermal energy = $mc\Delta\theta$
- specific latent heat: $E = mL$
- gas pressure comes from particle collisions
- at constant temperature: pV = constant

2.1 Particle model and internal energy

Q080 • Feb/March 2021 Paper 42 Question 4 — 2.1 Particle model and internal energy

(a) molecules strike walls

momentum (of molecules) changes / momentum = mass × velocity

force = rate of change of momentum

pressure = (sum of) force(s) / area / pressure = rate of change of momentum / area

(b)(i) (p2=) p1V1 / V2

p1V1 = p2V2

(b)(ii) greater

molecules move faster / have greater KE / molecules have greater momentum

(leads to) more frequent / harder collisions (with walls) / great rate of change of momentum

Q081 • May/June 2021 Paper 41 Question 5 — 2.1 Particle model and internal energy

(a) molecules / they speed up or gain kinetic energy

molecules move further apart or push others away

(b) forces between liquid molecules weak(er than in solids)

less energy / work done to separate molecules; or greater separation for same work done / same increase in energy

(c)(i) greater sensitivity

volume increase (of liquid in second thermometer) is greater

or liquid moves a greater distance (for the same temperature increase)

(c)(ii) smaller range and either of:

• smaller temperature increase for liquid / meniscus to reach end of tube

• expands more / greater sensitivity and tube of same length

(d)(i) statement of problem (e.g. bridges buckle (in hot weather))

(d)(ii) suggested solution to problem stated in 5(d)(i) (e.g. gaps at the ends of the bridge)

more detail (e.g. as the bridge expands the gaps close)

Q082 • Oct/Nov 2021 Paper 41 Question 3 — 2.1 Particle model and internal energy

(a) any three of:

they / molecules collide with inner surface

momentum (of a molecule) changes / reverses

force exerted / impulse

force spread over area / surface or p = F / A

(b)(i) (V2 =) p1V1 / p2 in any form or 630 × 1.0 × 105 / 1.4 × 105

450 cm3 or 4.5 × 10-4 cm3 or 0.45 dm3

(b)(ii) any two of:

molecules move more slowly / have less kinetic energy

pressure (inside balloon) decreases or pressure is directly proportional to temperature or p ∝ T

volume is directly proportional to temperature or V ∝ T

molecular collisions less frequent

molecular collisions less violent / hard / exert smaller impulse

water / external pressure compresses balloon or

water pressure greater (and balloon compressed)

Q083 • Oct/Nov 2021 Paper 43 Question 4 — 2.1 Particle model and internal energy

(temperature of air increases) so molecules move faster / their KE increases

molecules collide with walls of container and change momentum

greater change of momentum when temperature is higher or collisions more frequent or harder collisions or force = rate of change of momentum

(higher force and hence) higher pressure

Q084 • Feb/March 2022 Paper 42 Question 3 — 2.1 Particle model and internal energy

(a) Any two valid point(s):
- (Amount of water in the pool decreases) as water evaporates / becomes water vapour / gas
- The (more) energetic molecules escape; or fast(er) molecules escape
- From the surface of the water

(b)
lower temperatures / cold(er) day or less windy weather
produces smaller pool more slowly as a draught over
produces smaller pool more slowly because rate of
or surface removes water vapour enabling faster rate of
evaporation decreases with decreasing temperature
evaporation

(c) Any three valid point(s):
- (thermal) energy in the skin / body transferred to (molecules of) sweat
- These molecules (have enough KE to) escape from the skin / become water vapour
- Leaving behind molecules with lower energy
- Which leaves the skin / body at a lower temperature

Q085 • May/June 2022 Paper 43 Question 4 — 2.1 Particle model and internal energy

(a)(i) zig zag motion / random changes of direction
random length of path in each direction

(a)(ii) Any four valid point(s):
- air molecules bombard smoke particles
- air molecules are small (compared to smoke particles) / have small(er) mass
- air molecules are very fast moving
- air molecules move in random directions
- (collisions exert unbalanced) forces on smoke particles

(b)(i) kinetic energy (and potential energy) of molecules increase (hence internal energy increases)

(b)(ii) bigger change in momentum of molecules or molecules hit (the walls) harder
(molecules hit) more often / more frequently

Q086 • Oct/Nov 2022 Paper 42 Question 4 — 2.1 Particle model and internal energy

(a) Any three valid point(s):
- moving particles have momentum or particles hit walls
- momentum changes when particles hit walls
- force exerted (by particles) due to (rate of) change of momentum
- pressure is (total) force (of particles) per unit area (of wall).

(b) pressure increases
(there is a) greater change of momentum; or (particles exert) greater force (on same area)
particles have more KE

(c) (pressure =) 1.5 × 105 Pa
p1 V1 = p2 V2 or (p2 =) p1 V1 /V2 or pV = constant (for fixed m, fixed T)
(p2 =) 9(.0) × 104 × 170 / 100

Q087 • May/June 2023 Paper 41 Question 3 — 2.1 Particle model and internal energy

(a) heated / hot(ter) / warm(er) air is less dense or cool(er) air is more dense
heated / hot(ter) / warm(er) air rises (to ceiling displacing cooler air); or cool(er) air falls (displaced by warm(er) air)

(b)(i) speed / velocity (of particles) increases or (they) move faster

(b)(ii) (higher temperature means) particles collide (with rubber) harder / with more force / with greater momentum (change)
(larger volume means) particles collide (with rubber) less frequently

(larger volume means) larger (surface) area (for particle collisions)
effect of larger volume cancels effect of increased temperature /
the effect of larger area cancels the effect of larger force /
P = F / A so the two changes cancel each other /

Q088 • Oct/Nov 2023 Paper 41 Question 3 — 2.1 Particle model and internal energy

(a) particles (of liquid) are touching / close to each other
forces (of repulsion) between particles (of liquid) are large
(b)(i) (Δp =) ρ g(Δ)h
1000 × 9.8 × 0.087 or (Δp =) 852.6 (Pa)
(b)(ii) 12 N
p = F / A or (F =) pA or 850 × 0.014
(b)(iii) 1.2 kg
g = W / m or (m =) F / g or 12 / 9.8

Q089 • Oct/Nov 2023 Paper 43 Question 4 — 2.1 Particle model and internal energy

(a)(i) absolute zero
(a)(ii) -196 (°C)
(absolute zero / 0 K) = -273 (°C)
(b) increased (pressure)
particles of gas move faster or have more KE / momentum / velocity
any two from from:
• more frequent collisions of particles (with walls)
• particles collide (with walls) with a larger force or larger impulse
• (greater change in momentum of particles) causes greater force (on walls)
• pressure = force / area
(c) (pressure =) 60 kPa or 6.0 × 104 Pa
p1 V1 = p2 V2 or (p2 =) p1 V1 / V2 or pV = constant
or p ∝ 1 / V or 1.2 × 105 × 0.5 or 6.0 × 10n

Q090 • May/June 2024 Paper 43 Question 5 — 2.1 Particle model and internal energy

(a) Any three valid point(s):
• (metals contain) delocalised / free / mobile electrons
• (delocalised electrons) gain energy (from lattice vibrations of the atoms nearest to the hot water)
• (delocalised) electrons move through the metal (lattice)
• collisions between the (delocalised) electrons and (remote) ions / atoms (transfers thermal energy to all parts of the metal container)
(b) no delocalised / free electrons
(c) particles are further / far apart
fewer particle collisions (to transfer energy) or no lattice vibrations (to transfer energy)

Q091 • Oct/Nov 2025 Paper 43 Question 4 — 2.1 Particle model and internal energy

(a)(i) 930 kg / m3
(a)(ii) 1400 N
(Δ)p = ρg(Δ)h or 930 × 9.8 × 1.1 or 10 000 (N / m2)
area = length × width or 0.45 × 0.30 or 0.14 (m2)
(F=) p × A or 10 000 × 0.14
Alternative method:
1400 N
volume (of liquid above block) = length × width × height or 0.45 × 0.30 × 1.1 or 0.15 (m3)
mass (of liquid above block) = ρ × v or 930 × 0.15 or 140 (kg)
(F =) mg or 140 × 9.8

(b) thermal expansion
or to prevent buckling / bending when heated
Any one valid point(s):
- particles vibrate faster
- particles have greater speed / kinetic energy / momentum
- vibrations (of particles) take up more space
- (average) space / separation between particles increases

(c) (plastic is a good) insulator / poor conductor
(handle) will not get too hot to touch
or reduces heat flow to hand

2.2 Gas pressure and kinetic model

Q092 • May/June 2021 Paper 42 Question 5
2.2 Gas pressure and kinetic model

(a) air good insulator / poor conductor
holder / it stops / reduces conduction or
no / less thermal energy conducted (to hand)
temperature (of outside of holder) lower (than cup); or less energy to skin / hand / person
(b) (put a) lid / cover (on cup)
mention of convection
less / no convection (from surface)
alternative route for last 2 m.p.s
mention of evaporation
less / no evaporation (from surface / container)
(c) radiation

Q093 • Oct/Nov 2021 Paper 43 Question 3
2.2 Gas pressure and kinetic model

(a)(i)
pressure in a liquid increases with depth or pressure decreases (as bubble rises)
pressure (of gas) is inversely proportional to volume; or internal pressure greater than external pressure (momentarily)
Key physics: air molecules do not have hit surface bubble as frequently stop collapsing
as strongly compressed
(a)(ii) 0.50 cm3
PV = constant, in any form
P (due to water) = ρgh, in any form
[1.0 × 105 + (1000 × 10 × 3.0)] × 0.40 = [1.0 × 105 + (1000 × 10 × 0.5)] × V2
(b)
paper is not compressed as much / less force on piston B
air can be compressed; or some of the energy is used to compress the air (instead of the paper)

Q094 • Oct/Nov 2022 Paper 41 Question 4
2.2 Gas pressure and kinetic model

(a)(i) 240 N
F = pA in any form or 1.0 × 105 × 2.4 × 10-3
(a)(ii) 5.0 J
WD = Fx parallel or 240 × 0.021
(b) (-)3.5 × 103 J
E = CDT in any form or 89 × (21 - (-18) or 89 × (3) or 89 × 39
(c)
(as the volume decreases) the particles collide more often
(as the temperature decreases) the particles collide less violently
two effects cancel (to leave the pressure unchanged); or particles collide with walls / piston / cylinder

(d)
(attractive) forces between (any two) particles large(r than in gases)
particles close(r) together (than gas particles) or particles already touching

Q095 • Oct/Nov 2022 Paper 42 Question 2 2.2 Gas pressure and kinetic model

(a) (use stop-watch to) time oscillations
(use of fiduciary) aid to determine a complete cycle
(use of) multiple oscillations and division (to determine period)
(b)
quantity device
volume of water in a glass measuring cylinder
width of a small swimming pool metre rule
thickness of a piece of aluminium foil micrometer screw gauge
1 mark for each correct response

Q096 • May/June 2023 Paper 42 Question 4 2.2 Gas pressure and kinetic model

(a) pressure decreases and particles have smaller velocity / momentum / smaller EK / kinetic energy (when temperature is
lower)
lower rate / frequency of collision of particles
particles collide with smaller force or smaller impulse change
(b)(i) -273 (°C)
(b)(ii) (temperature at which) particles have least EK / kinetic energy
lowest possible temperature
(c) 200 cm3
pV = constant or 9.0 × 104 × 350 = 1.6 × 105 × V2
V2 = [9.0 × 104 × 350] / 1.6 × 105 or V2 = 2.0 × 10N or 1.97 × 10N

Q097 • Oct/Nov 2025 Paper 42 Question 4 2.2 Gas pressure and kinetic model

(a)(i) 170 (cm3)
pV = constant or 1.0 × 105 × 240 = 1.4 × 105 × V2
(V2 =) {1.0 × 105 × 240} / 1.4 × 105 or 1.7 × 10N
(a)(ii) 270 N
(F =) pA or (F =) 1.4 × 105 × 1.9 × 10-3
(a)(iii) (W =) Fd in words or symbols
7.9 J
(b) distance between gas particles is much larger (than distance between liquid particles)
or particles in liquid are touching and particles in gas are far apart

2.3 Specific heat capacity

Q098 • Oct/Nov 2021 Paper 43 Question 5 2.3 Specific heat capacity

(a)(i) 1.2 kg
7600 × 0.41
(m =) volume constant so mass directly proportional to density
(a)(ii) 0.37 J / °C
(thermal capacity =) mass × specific heat capacity
(a)(iii) 48 J
(E =) mcΔT or 1.2 × 0.50 × (100 - 20) in any form
(b) electrons mentioned
(metals have) electrons free to move / delocalised (which transfer thermal energy)

Q099 • Feb/March 2022 Paper 42 Question 4 — 2.3 Specific heat capacity

(a)(i) (mass = 1900 × 0.05) = 95 kg

(m =) ρV in any form or 1900 × 0.05

(a)(ii) (= 95 × 1500) = 140 000 J / °C / 1.4 × 105 J / °C

(C =) m × c

(a)(iii) 34 000 s / 560 min / 9 h 20 min

Temperature rise = 12(°C) / 19-7

E = C × Δ θ in any form

(t =) E / P in any form or (thermal capacity × 12) / 50

(b)(i) sand molecules gain KE or vibrate more or hit (other) molecules (when heated)

(thermal energy is transferred by) conduction

Energy is transferred to molecules of plastic pot in contact with sand (through collisions)

Energy or (lattice) vibrations transferred to neighbouring molecules

(b)(ii) (sand is warmer than surroundings and so thermal) energy (constantly) is lost from the sand

Key physics: constant temperature rate thermal energy supplied sand equal lost

Q100 • Oct/Nov 2022 Paper 42 Question 5 — 2.3 Specific heat capacity

(a) (E =) 410 000 000 J or 410 MJ or 4.1 × 108 J

ΔE = mcΔT or (ΔE =) mcΔT or 1200 × 960 × 360

(ΔT =) 360 (°C)

(b) (thermal) radiation

electromagnetic / e-m / infrared / IR (radiation emitted from block)

travels to worker or is absorbed by worker or travels without needing a medium

(c) conduction

delocalised / free / moving electrons

Any one valid point(s):

- (electrons) move (from outer surface) to interior (of rollers)
- (electrons) travel through(out) the solid / large distances
- (electrons) collide with distant particles
- lattice vibrations transfer thermal energy to neighbouring particles; or particles vibrate and cause nearby / adjacent particles to vibrate or vibrating particles collide with particles transferring energy.

Q101 • May/June 2023 Paper 42 Question 5 — 2.3 Specific heat capacity

(a)(i) ρ = m / V or m = ρ V

(m =) 1.2 × 4.5 × 6.1 × 2.4 (= 79 kg) or (m =) 79.056 (kg)

(a)(ii) 290 s

c = (Δ)E / mΔθ or (ΔE =) mcΔθ or (ΔE =) 79 × 1000 × 4(.0) or (ΔE =) 316 000 or (Δθ =) 4(.0)

P = (Δ)E / t or (ΔE =) Pt or (ΔE =) 1100 × t

(t =) mcΔθ / P or (t =) 79 × 1000 × 4(.0) / 1100 or (t =) 316 000 / 1100

(a)(iii) Any one valid point(s):

- (thermal) energy is transferred to furniture / walls / objects (in the room)
- (thermal) energy is transferred through windows / doors / floor / ceiling / from the room

(b) conduction and convection

Q102 • May/June 2023 Paper 43 Question 3 — 2.3 Specific heat capacity

(a)(i) (greatest) gas

(least) solid

(a)(ii) Any three valid point(s):

- (solids) particles vibrate
- (gases) particles move freely
- (solids) particles in fixed / close positions

• (gases) particles randomly arranged (in container) / wide separation
• (gas) particles move quickly

(b)(i) (specific heat capacity is the) energy required to raise 1 kg / unit mass by 1 °C / 1 K / 1 kelvin / unit temperature

energy required per unit mass / 1 kg

or energy required per unit temperature / 1 °C increase / 1 K increase / 1 kelvin increase

(b)(ii) time

mass

initial temperature and final temperature

Q103 • Oct/Nov 2023 Paper 42 Question 4 — 2.3 Specific heat capacity

(a)(i) Any three valid point(s):

• increase in the (average) KE / speed of air particles
• more frequent collisions of (air) particles (with bottle)
• more forceful collisions of (air) particles (with bottle)
• greater force per unit area gives greater pressure
• volume unchanged and so pressure increases

(a)(ii) (pressure decreases as)

air (particles) escape from the bottle / into the air

Key physics: until pressure inside bottle same as air outside

atmospheric pressure

(b) 1.5 × 104 J

c = (Δ)E / mΔθ (ΔE =) mcΔθ or (ΔE =) 0.18 × 4200 × 20

(c) 3900 Pa

(Δp =) ρg(Δ)h or (Δp =) 1.0 × 103 × 9.8 × 0.4 or

(Δp =) 1.0 × 103 × 9.8 × 40 or (Δp =) 3.9 × 10N

Q104 • Feb/March 2024 Paper 42 Question 4 — 2.3 Specific heat capacity

(a) energy transferred per unit mass per unit temperature change

(thermal) energy (transferred) per unit temperature change

(b)(i) (m =) 2.2 kg

m = ρV in any form or 910 × 0.0024

(b)(ii) 7.0 × 105 J or 700 000 J

c = ΔE / { mΔθ } or (ΔE =) mcΔ θ or 2.2 × 2000 × 160 or 2.184 × 2000 × 160

(b)(iii) 1700 W

(P =) E / t or 700 000 / 7 × 60 or 704 000 / 7 × 60 or 698 880 / 7 × 60

Q105 • May/June 2024 Paper 41 Question 3 — 2.3 Specific heat capacity

(a) (they / particles in ice) vibrate (about a fixed position); or particles in water move throughout the liquid

(b)(i) conduction

(b)(ii) 2.6 × 104 J

c = (Δ)E / mΔθ or (ΔE =) mcΔθ or 0.34 × 4200 × 18 or 2.6 × 10N (J)

(b)(iii) density (of water next to the ice) increases

cold(er) water sinks

warm(er) water replaces cold water or warm(er) water rises or making a convection current

(b)(iv) internal energy decreases and (average) kinetic energy (of particles) decreases

kinetic energy decreases

Q106 • May/June 2024 Paper 43 Question 4 — 2.3 Specific heat capacity

(a) (average) KE of particles increases / particles move faster

(b)(i) ρ = m / v or (m =) ρv

1 cm3 = 1 × 10-6 m3 or 250 cm3 = 2.5 × 10-4 m3
or 1000 × 2.5 × 10-4 (= 0.25 kg)
(b)(ii) 47000 J
(Δθ =) 65 - 20 °C or (Δθ =) 45 °C
E = mcΔθ or (E =) mcΔθ or (E =) 0.25 × 4200 × 45
(b)(iii) Key physics: thermal energy also transferred pan / surroundings escapes water as heated
(c) Any two valid point(s):
- (aluminium saucepan) takes longer to heat the water
- more (thermal) energy is needed (with aluminium pan for the same increase in temperature)
- (because aluminium) has a higher specific heat capacity

Q107 • Oct/Nov 2024 Paper 43 Question 3 — 2.3 Specific heat capacity

(a)(i) good/better absorber (of radiation) or bad / poor / worse reflector (of radiation)
(a)(ii) doesn't work at night / in cloud cover / when there is no sun
or (sun has) variable output
(b) Any two valid point(s):
- hydroelectric
- tidal
- wave
- wind
- geothermal
- biofuels

(c) 9.2 × 106 J or 9 200 000 J
(temperature rise =) 33 (°C) or 43 - 10
c = ΔE / mΔθ or (E =) m × c × Δθ or (E =) 40 × 4200 × (43 - 10) or 5.5 × 106 (J)
5.5 × 106 × (100 / 60) or 5.5 × 106 × 1.667 or 5.5 × 106 / 0.6
useful energy output
or efficiency = × 100%
total energy input

Q108 • May/June 2025 Paper 43 Question 4 — 2.3 Specific heat capacity

(a) Any three valid point(s):
- (copper / metal contains) free / delocalised electrons
- electrons carry (thermal) energy through metal
- electrons collide with (distant) ions
- lattice vibrations transfer energy (to neighbouring ions); or ions vibrate and cause (nearby / adjacent) ions to vibrate

(b)(i) energy transferred per unit mass per unit temperature change
(thermal) energy (transferred) per unit temperature change
(b)(ii) 0.51 J / (g °C)
(energy lost by metal =) 54 × c × 69
or (energy gained by water =) 50 × 4.2 × 9
or 1890
energy lost by metal = energy gained by water
or 54 × c × 69 = 50 × 4.2 × 9

Q109 • Oct/Nov 2025 Paper 41 Question 4 — 2.3 Specific heat capacity

(a) 24 000 J or 2.4 × 104 J
(ΔE =) mcΔθ or (ΔE =) 0.15 × 4200 × {58 - 20}
(Δθ =) 58 - 20 or 38 (°C) seen
(b) (temperature of point X) on metal rod increases faster
(temperature of point X) increases (with time)
thermal energy is transferred) by conduction

metal rod (transfers thermal energy through movement of) delocalised / free electrons.

2.4 Specific latent heat and changes of state

Q110 • May/June 2021 Paper 41 Question 4 — 2.4 Specific latent heat and changes of state

(a) aluminium is a (good) conductor (of heat) and plastic is a poor conductor / does not conduct (heat)

(b)(i) increase in kinetic energy of molecules or increase in potential energy of molecules

(b)(ii) Any three valid point(s):

- atoms (touching the hotplate) / lattice vibrate (faster)
- atoms pass on energy / vibration to neighbouring atoms / to other atoms by collision
- atoms pass on energy to electrons
- electrons hit distant atoms or electrons move (through lattice)

(b)(iii) molecules escape from the liquid (as a vapour)

bonds broken / (attractive) forces overcome

molecules gain potential energy

or work done (to separate molecules / break bonds / overcome forces)

(b)(iv) 840 W

(E =) mlv in any form or 0.11 × 2.3 × 106 or 2.53 × 105

(rate =) mlv / t in any form or 0.11 × 2.3 × 106 / 300 or 2.53 × 105 / 300

2.5 Evaporation

Q111 • May/June 2021 Paper 42 Question 4 — 2.5 Evaporation

(a)(i) random / haphazard / zig-zag / irregular

(a)(ii) (liquid / water) molecules move fast or (pollen) particles massive

collide / bombard

uneven collisions / collisions from different directions (cause random movement); or (liquid / water) molecules move randomly

(b)(i) cooling

(thermal) energy used / needed to evaporate (ethanol) / overcome attractive forces(between molecules / particles)

thermal energy taken from skin / patient / person

alternative route for last two m.p.s

more / most energetic (liquid) molecules / particles escape; or less / least energetic (liquid) remain

less / least energetic molecules / particles linked to lower temp (of skin)

(b)(ii) greater / increases / faster / higher

Q112 • May/June 2022 Paper 41 Question 3 — 2.5 Evaporation

(a) fast(er) / high(er) speed / (more) energetic molecules escape (into air)

average speed / average kinetic energy of molecules decreases

temperature related to speed / energy of molecules; or slow(er) / low(er) speed / less energetic molecules remain (in water)

(b) Any three valid point(s):

atoms / ions vibrate

(vibrating) atoms / ions hit electrons

electrons propelled / travelling through metal / moving through metal

electrons hit (distant) atoms

free electrons / delocalised electrons mentioned

Q113 • Oct/Nov 2022 Paper 43 Question 5 — 2.5 Evaporation

(a) energy from the Sun transfers to / is absorbed by (water) molecules, (so KE of (water) molecules increases)

molecules with high(er) energy / KE / fast(er) moving molecules escape (from the surface)
wind removes molecules when they have left the surface (so they do not re-enter the liquid)
Any one valid point(s):

- wind increases the rate of evaporation
- (absorption of) energy from the Sun increases the rate of evaporation
- least / less water evaporates / lower rate of evaporation from dish C
- most / more water evaporates / higher rate of evaporation from dish B

(b) energy to change 1 kg / unit mass from liquid to gas / gas to liquid (without changing its temperature)
energy to change from liquid to gas / gas to liquid
or energy to change state of 1 kg
(c) A: temperature (of solid / ice) increases and C: temperature (of liquid / water) increases
B: solid / ice changes to liquid / water; or solid / ice melts (at constant temperature)
D: liquid / water changes to gas / steam; or liquid / water boils (at constant temperature)

Q114 • Feb/March 2023 Paper 42 Question 4 — 2.5 Evaporation

(a) delocalised / free / mobile electrons
electrons move through metal; or electrons collide with distant particles
lattice vibrations transfer energy to neighbouring particles; or particles vibrate and cause nearby / adjacent particles to vibrate or vibrating particles collide with particles transferring energy
(b)(i) (attractive) forces (between particles are much) greater in liquids (than in gases)
particles in gases are (much) further apart (than in liquids)
(b)(ii) occurs at a fixed temperature
takes place throughout the liquid
(c) 3.93 J / (g °C) or 3930 J / (kg °C)
$\rho = m / V$ or ($m =$) ρV or 1.03×200 or 206 SEEN
$c = E / m\Delta\theta$ or ($c =$) $E / m\Delta\theta$ or $60\,700 / (206 \times 75)$ or $60700 / (1.03 \times 200 \times 75)$
($m =$) 206 (g) or ($\Delta\theta$) = 75 (°C)

Q115 • May/June 2024 Paper 42 Question 4 — 2.5 Evaporation

(a) (evaporation:) (only) at the surface or
boiling: happens throughout the liquid
(evaporation:) takes place at any temperature or
boiling: takes place at a specific temperature / boiling point
(b)(i) 113 (K)
(b)(ii) conduction
convection
(c) particles collide with the walls / container
(particles) exert a force on the walls; or collision with walls produces a change in momentum (of particles)
pressure is force per unit area or $p = F / A$ or pressure is rate of change of momentum per unit area

Q116 • Oct/Nov 2024 Paper 41 Question 4 — 2.5 Evaporation

(a) 1 Any one method to transfer measurable amount of thermal energy for $\Delta\theta$:
(a) to aluminium block (with electrical heater)
(b) from aluminium block to known liquid
(c) from known liquid to insulated aluminium (calorimeter)
(d) to known liquid and aluminium (calorimeter)
2 Determination of energy transferred for $\Delta\theta$, to match workable method in 1:
(a) Use of $E = Pt$ or $E= IVt$
(b) Use of $E = mc\Delta\theta$ with s.h.c. of known liquid
(c) Use of $E = mc\Delta\theta$ with s.h.c. of known liquid
(d) Use of $E = Pt$ or $E = IVt$ and $E = mc\Delta\theta$ (with known s.h.c. of liquid)
3 Any one measurement from:

• initial and final temperature / temperature change
• time (of heating)
• mass of aluminium
c = E / mΔθ or (c =) E / mΔθ
(b)(i) Any three valid point(s):
1 (net) transfer of energy from higher temperature to lower temperature or (net) transfer of energy from water / to dish
2 (energy transfer) by conduction or aluminium is a good conductor (of thermal energy)
3 temperature of water decreases and temperature of dish increases
no (net) transfer of energy when temperature of dish = temperature of water
(b)(ii) (particles) gain energy in kinetic store (as temperature of aluminium increases)
(average) separation of (aluminium) particles increases or (aluminium) particles move further apart
(b)(iii) (water) molecules with more/enough energy escape from the surface
escape of more energetic molecules (from water) or (molecules) leave from the surface

Q117 • Oct/Nov 2024 Paper 42 Question 4 — 2.5 Evaporation

(a) boiling happens at a specific temperature; or evaporation happens at a range of temperatures
any temperature (below the boiling point)
evaporation happens at the surface of the water or boiling happens throughout the water
(b) Any four valid point(s):
1 (as the water is heated) the number of gas particles increases
2 (particles) gain internal / kinetic energy
3 (there are) more frequent collisions between particles and surface / wall / lid of the cooker
(each) collision (of particles) is harder / exerts more force (on cooker surface); or greater change of momentum when
particles collide (on cooker surface)
5 pressure ∝ (total) force (of collisions) or pressure = force / area

Q118 • Oct/Nov 2025 Paper 42 Question 5 — 2.5 Evaporation

(a)(i) particles with more (kinetic) energy escape or fast(er) particles escape
(particles) escape from the surface
(a)(ii) windy or (air) temperature higher / Sun shining
particles that escape from surface blown away (and unable to return)
or more energy given to the particles (to escape)
(b) braking distance increases and friction (between tyres and road) decreases

2.6 Thermal physics - mixed

Q119 • Feb/March 2021 Paper 42 Question 3 — 2.6 Thermal physics - mixed

(a) renewable / yes
crops can be regrown (to replace resource) / waste materials don't run out
(b) water will cool (too much) / thermal energy lost (during transfer)
lag/insulate (pipes) or transport in a poor conductor of thermal energy
(c) Any two valid point(s):
• air pollution / harmful gases / acid rain
• CO2 / greenhouse gases / contribution to global warming
• not renewable
• damage from mining / drilling or any valid environmental consequence of transport of coal

Q120 • Oct/Nov 2021 Paper 42 Question 5 — 2.6 Thermal physics - mixed

(a) wires of 2 different metals
one junction clearly in each liquid

voltmeter / ammeter / galvanometer correctly connected

(b) any two from

- expansion of liquid
- expansion of solid
- expansion of gas
- density (of liquid)
- (electrical) resistance

(c) any two from

- large range
- (measure) high temperatures
- remote sensing
- small size or small mass
- small thermal capacity
- suitable for data logging
- responds quickly or measures rapidly varying temperatures or temperature changing continuously

Q121 • May/June 2022 Paper 41 Question 4

2.6 Thermal physics - mixed

(a)(i) two / three wires of at least two different metals

one junction in sulfur

Key physics: other junction ice-water mixture / room temperature wires must first

labelled voltmeter / voltmeter symbol correctly connected

(a)(ii) measure e.m.f.

how to find temperature from e.m.f. (e.g. use calibration graph or calculation or table)

(b) Key physics: measures high temperatures / wires do not melt rapid robust small heat capacity electrical output can

remote from observer / direct input to computer

Q122 • May/June 2022 Paper 42 Question 4

2.6 Thermal physics - mixed

(a) statement: bore of constant (cross sectional) area

explanation: idea of same movement / change in length of liquid / thread

for same increase in volume / expansion (of liquid)

statement: (liquid has) constant thermal expansion

explanation: liquid moves same distance for each °C temperature rise

(b) heat capacity / it is small

only uses / needs a small amount of (thermal) energy (to raise its temperature)

(c) 36 J

(E =) CΔT in any form

(E =) 0.11 × (345 - 20) or (ΔT =) 325 (°C)

Q123 • Oct/Nov 2022 Paper 43 Question 4

2.6 Thermal physics - mixed

(a)(i) Any one valid point(s):

- volume (of liquid)
- length (of thread / liquid in tube).

(a)(ii) more or greater (sensitivity)

volume of liquid / length of thread increases more per °C / unit temperature (because greater volume of liquid present)

Key physics: more liquid expand so gives larger change level per °C / unit temperature

(a)(iii) longer (capillary) tube

liquid can expand further so to a higher temperature

smaller (volume) bulb

less liquid so liquid expands less / lower rise per °C

larger diameter / wider capillary tube

lower increase in level for each °C

replace liquid with a liquid with lower expansivity
liquid expands less for each °C
(b) e.m.f.

Q124 • Oct/Nov 2023 Paper 41 Question 2 — 2.6 Thermal physics - mixed

(a) Any three valid point(s):
• free / delocalised / mobile electrons
• (electrons) gain (thermal) energy from hotplate / particles
• (electrons) move through(out) copper / metal or (electrons) move to distant particles
• electrons transfer energy from higher temperature (region) to lower temperature (region) or (electrons) collide with (distant) particles / transfer energy to (distant) particles
(b) (shiny surfaces are) poor emitters of radiation
reduces energy loss (from the pan / copper) or less energy transferred to surroundings
(c) convection

Q125 • Oct/Nov 2024 Paper 42 Question 5 — 2.6 Thermal physics - mixed

(a) (tarmac / it) absorbs infrared radiation (emitted from the Sun)
(tarmac / it) absorbs radiation / infrared (emitted from the Sun)
(b) conduction
(c) convection
(d) Any two valid point(s):
1 black / tarmac is a better absorber (of radiation) than air
2 tarmac is a poor emitter (at low / this temperature)
3 thin layer of tarmac / very large volume / column of air above road

Q126 • Oct/Nov 2025 Paper 43 Question 3 — 2.6 Thermal physics - mixed

(a)(i) (wasted power =) 2200 + 40 + 25 or 2265
or (useful power output =) power input - wasted power
1140 (MW) (≥ 3 sf)
(a)(ii) 0.32 or 32%
(efficiency =) {useful power output / total power input} (× 100%)
or 1100 / 3400
(b) water is pumped into the ground and heated by hot rocks
heat from hot rocks underground is used to turn water into steam
steam used to heat buildings
steam used to turn generator / turbine / produce electricity